

transmission proportional to size of within-patch epidemic

At the beginning of the season if the proportion of infected patches is p and population density is N , then after this season only $P_1 p$ persists.

During the season some patches get infected because of transmission. We assume this amount is proportional to the size of within-patch epidemic z (the final size) .

It is also proportional to amount of currently infected patches $P_{nf} p$, where P_{nf} is the non-fizzle probability (fizzling patches not included because they are not going to spread disease to other patches) and proportional to amount of currently susceptible patches $1 - p$.

So at the beginning of next season p becomes $P_1 p + az P_{nf} p(1 - p)$, a is a scaling constant in case it's useful. We can also use hazard model for this version, i.e., $P_1 p + (1 - e^{-az P_{nf} p})(1 - p)$.

The change of N is the same as previous version: $N \rightarrow (1 + r)(1 - zD)N + c(1 - p)(S - (1 + r)(1 - zD)N)$.

So the overall function is

$$f : \begin{pmatrix} p \\ N \end{pmatrix} \rightarrow \begin{pmatrix} P_1 p + (1 - e^{-az P_{nf} p})(1 - p) \\ (1 + r)(1 - zD)N + c(1 - p)(S - (1 + r)(1 - zD)N) \end{pmatrix}$$

The disease-free equilibrium is $p = 0$ and

$$N_{DFE} = \frac{cS}{1 - (1 - c)(1 + r)(1 - zD)}$$

Local linearization:

$$J_f = \begin{pmatrix} P_1 + e^{-az P_{nf} p} az P_{nf} (1 - p) - (1 - e^{-az P_{nf} p}) & p \frac{\partial P_1}{\partial N} \\ -c(S - (1 + r)(1 - zD)N) & (1 - c(1 - p)(1 + r)(1 - zD)) \end{pmatrix}$$

At DFE it is

$$J_f = \begin{pmatrix} P_1|_{N=\frac{cS}{1-(1-c)(1+r)(1-zD)}} + az P_{nf} & 0 \\ -c(S - (1 + r)(1 - zD)N) & (1 - c)(1 + r)(1 - zD) \end{pmatrix}$$

So the persistence criterion is

$$L_0 > 1$$

where

$$L_0 = P_1|_{N=\frac{cS}{1-(1-c)(1+r)(1-zD)}} + az P_{nf}$$